Department of Computer Science
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SUBJECT NAME:	**COMPUTER ORGANIZATION AND ARCHITECTURE**
SUBJECT CODE:	**SECR 1033**
SEMESTER:	**2 - 2023/24**
LAB TITLE:	**Programming 1: Assembly Language Fundamentals**
INSTRUCTION:	**Student is required to have instructor's signature before proceed from each lab work.**

STUDENT INFO :

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SUBMITTED DATE: 23/4/2024

Lab # 1

Simple program to familiarize with Program Code, Rebuild & Start Without Debugging

Execute the programs below:
PART 1:

i. **Part A**: Adding and Subtracting Integers

```
TITLE Add and Subtract (AddSub.asm)

; This program adds and subtracts 32-bit integers
; Authors:
; Date:
; Revision:

INCLUDE Irvine32.inc

.data
TOTAL dword 0 ; a variable named TOTAL (declared as DWORD)

.code

main PROC

    mov eax, 123400h ; Set EAX with the value of 123400h
    add eax, 567800h ; Add the content of EAX with 567800h
    sub eax, 77700h ; Subtract content of EAX with 77700h
    mov TOTAL, eax  ; Store content of EAX to TOTAL

    call DumpRegs

    exit
main ENDP

END main
```

Experimental Results:
a) Rebuild & Start Without Debugging

Completed Fully Partially : Checked by: _____________

Screenshot Result:

```
 EAX=00613500  EBX=01038000  ECX=007710AA  EDX=007710AA
 ESI=007710AA  EDI=007710AA  EBP=012FFC5C  ESP=012FFC50
 EIP=00773679  EFL=00000206  CF=0  SF=0  ZF=0  OF=0  AF=0  PF=1


:\Users\User\source\repos\Lab Part A\Debug\Lab Part A.exe (proces
 16136) exited with code 0.
ress any key to close this window . . .
```

ii. **Part B**: Adding Variables

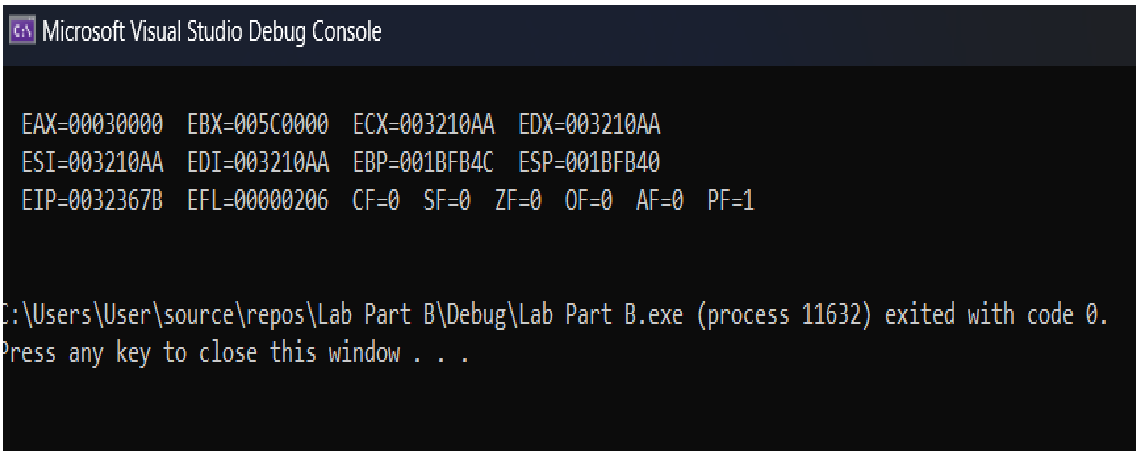
```
TITLE Add and Subtract, Version 2(AddSub2.asm)

; This program adds and subtracts 32-bit unsigned
; integers and stores the sum in a variable.
; Authors:
; Date:
; Revision:

INCLUDE Irvine32.inc
.data
val1 DWORD 10000h
val2 DWORD 40000h
val3 DWORD 20000h
finalVal DWORD ?
.code

main PROC
mov eax,val1      ; start with 10000h
add eax,val2      ; add 40000h
sub eax,val3      ; subtract 20000h
mov finalVal,eax ; store the result (30000h)

call DumpRegs     ; display the registers

exit

main ENDP
END main
```

Experimental Results:

a) Rebuild & Start Without Debugging

Completed *Fully* *Partially* *: Checked by: ______________*

Screenshot Result:



iii. **Part C**: Add and Subtract 8 and 16-Bit Version

```
TITLE Add and Subtract, Version 3
; This program adds and subtracts 8 and 16 bit
; unsigned integers and stores the sum in a variable.
; Authors:
; Date:
; Revision:

INCLUDE Irvine32.inc
.data
valw1 WORD 1000h
valw2 WORD 4000h
valw3 WORD 2000h
finalValw WORD ?

valb1 BYTE 10h
valb2 BYTE 40h
valb3 BYTE 20h
finalValb BYTE ?

.code
main PROC
mov ax,valw1; start with 10000h
add ax,valw2       ; add 40000h
sub ax,valw3       ; subtract 20000h
mov finalValw,ax   ; store the result (30000h)
call DumpRegs      ; display the registers

mov ah,valb1; start with 10000h
add ah,valb2       ; add 40000h
sub ah,valb3       ; subtract 20000h
mov finalValb,ah   ; store the result (30000h)
call DumpRegs      ; display the registers

exit
main ENDP
END main
```

Experimental Results:

a) Rebuild & Start Without Debugging

Completed   Fully   Partially   : Checked by: _____________

Screenshot Result:

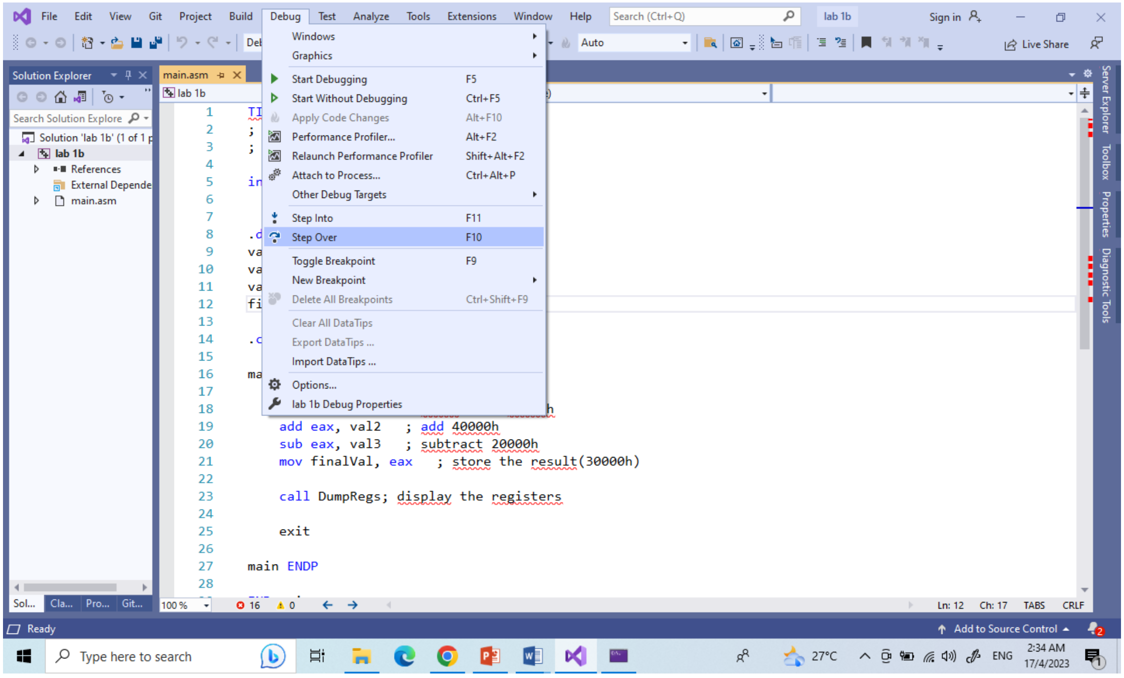
```
Microsoft Visual Studio Debug Console

 EAX=00933000  EBX=00721000  ECX=00E810AA  EDX=00E810AA
 ESI=00E810AA  EDI=00E810AA  EBP=0093FED8  ESP=0093FECC
 EIP=00E8367F  EFL=00000206  CF=0  SF=0  ZF=0  OF=0  AF=0  PF=1


 EAX=00933000  EBX=00721000  ECX=00E810AA  EDX=00E810AA
 ESI=00E810AA  EDI=00E810AA  EBP=0093FED8  ESP=0093FECC
 EIP=00E8369C  EFL=00000206  CF=0  SF=0  ZF=0  OF=0  AF=0  PF=1

:\Users\User\source\repos\Lab 1 Part C\Debug\Lab 1 Part C.exe (process 16804) exited with code 0.
ress any key to close this window . . .
```

Detail Debugging Process

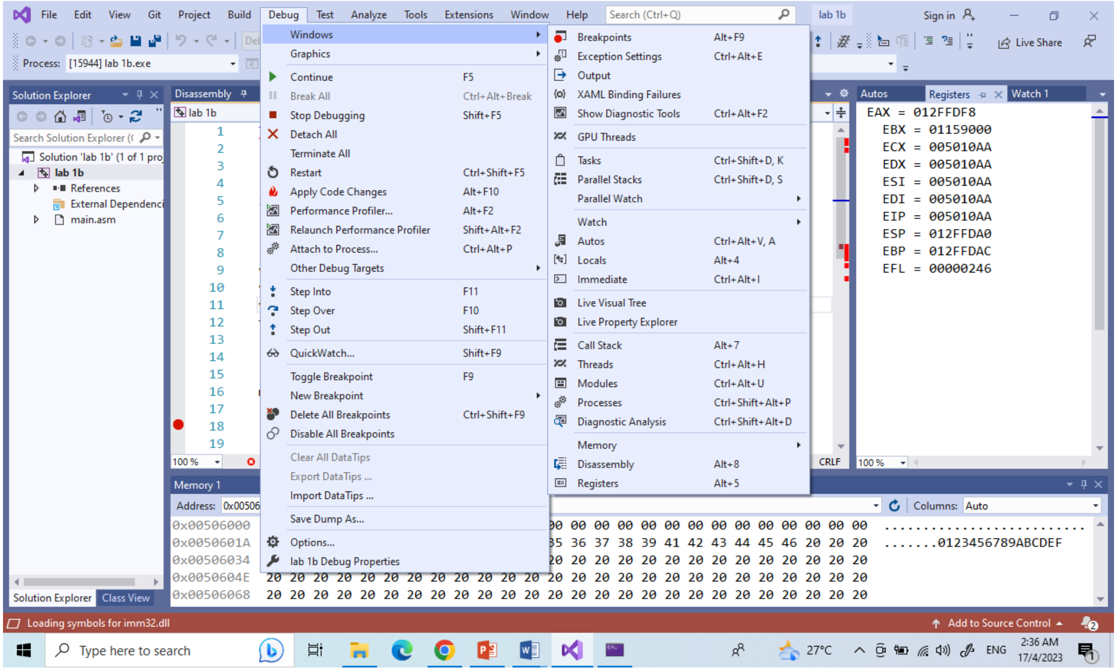
PART 2:

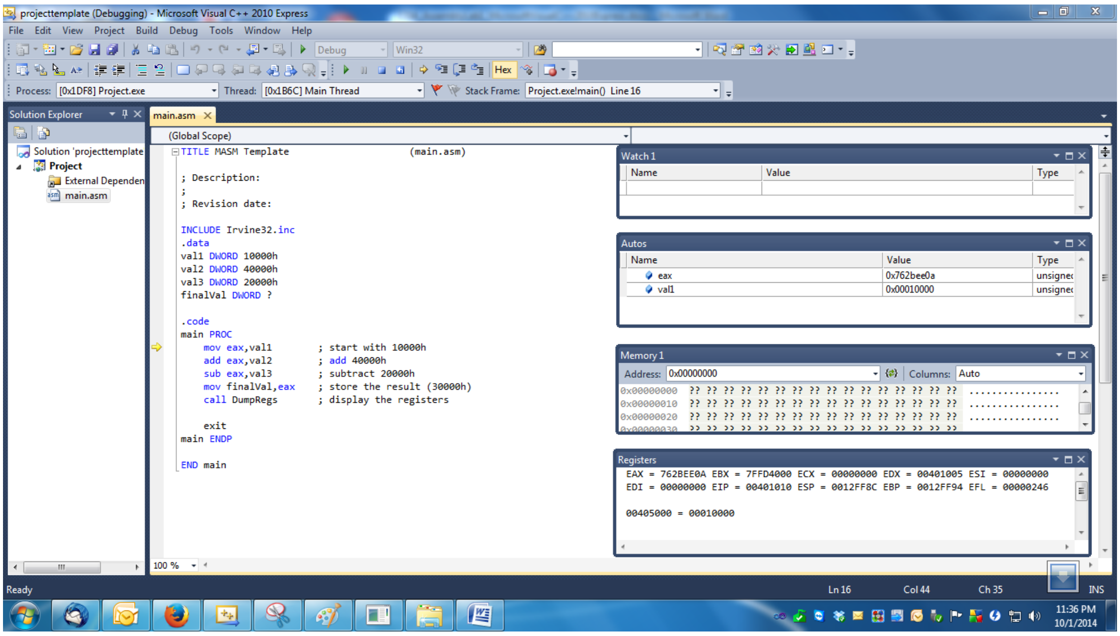
Run Debugging Process for Part B, Part C and Capture Video

1. Press F10 for step by step debugging.

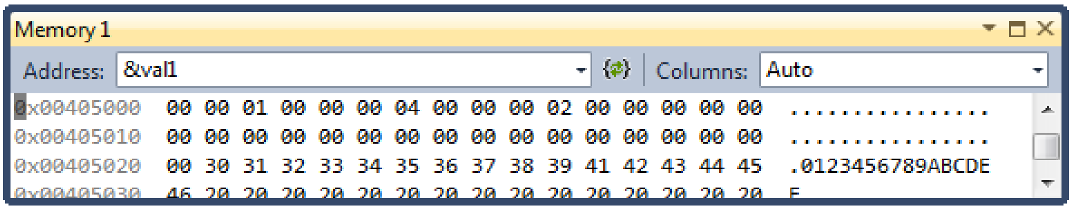


3. Open 5 windows:
    a. Watch
    b. Autos
    c. Memory
    d. Registers
    e. Disassembly





4. Default memory data segment value is at first variable: &val1 or 0x00405000 (depend on memory used).



5. Add variables val1, val2, val3 and finalval to Watch

Watch 1

Name	Value	Type
val1	0x00010000	unsigned long
val2	0x00040000	unsigned long
val3	0x00020000	unsigned long
finalVal	0x00000000	unsigned long

6. F10 to trace/step the assembly program line by line.

7. Please debug by looking at value changes at Watch, Memory, Autos and Registers windows.

8. The disassembly window can show the following optional information:

- Memory address where each instruction is located. For native applications, it is the actual memory address. For Visual Basic or C#, it's an offset from the beginning of the function.
- Source code from which the assembly code derives.
- Code bytes, that is, the byte representations of the actual machine or MSIL instructions.
- Symbol names for the memory addresses.
- Line numbers corresponding to the source code.

What is Byte Code (or Machine Code) for the following assembly instructions:

```
    mov ax,valw1  ; start with 10000h
    add ax,valw2        ; add 40000h
    sub ax,valw3        ; subtract 20000h
    mov finalValw,ax    ; store the result (30000h)
```

Byte Code:

```
mov ax,valw1     ; start with 1000h
00E83660  mov         ax,word ptr [valw1 (0E86000h)]
add ax,valw2           ; add 4000h
00E83666  add         ax,word ptr [valw2 (0E86002h)]
sub ax,valw3           ; subtract 2000h
00E8366D  sub         ax,word ptr [valw3 (0E86004h)]
mov finalValw,ax     ; store the result (3000h)
00E83674  mov         word ptr [finalValw (0E86006h)],ax
call DumpRegs    ; display the registers
00E8367A  call        _DumpRegs@0 (0E810E1h)
```

Experimental Results:

a) Rebuild & Start Debugging   CamStudio: 

Completed   *Fully*   *Partially*   *: Checked by: _____________*

Video submission :

https://drive.google.com/drive/folders/16ByK6W_7JMXezg7oE93yylvyC_rod
YX9?usp=sharing